

TCP HTCP Modified Metrics Analysis

Linear Change

The original CongestionAvoidance function typically has a more aggressive increase in cWnd using a quadratic scaling of m_alpha . Here is a less aggressive method using linear growth.

Modified Functions and Their Impacts:

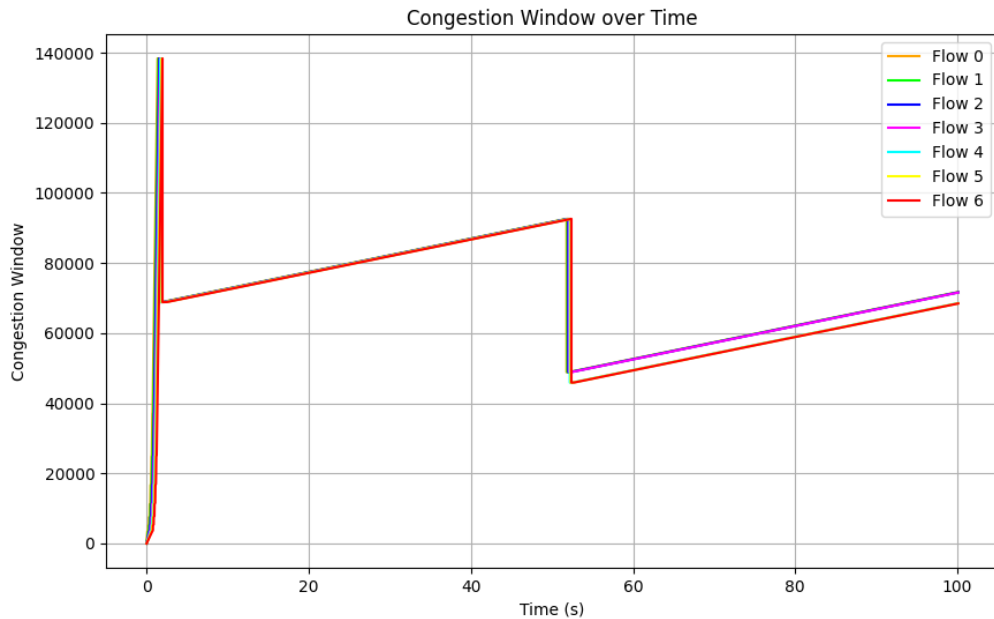
1. CongestionAvoidance Function

- **Change:**
 - Instead of quadratic or cubic growth, the congestion window increment (adder) is now **linear** with respect to m_alpha .
 - A lower bound of 1.0 is applied to ensure at least a minimal increase in cWnd, preventing it from stalling at low values.
- **Potential Impact:**
 - **Slower Window Growth:** The linear scaling reduces the aggressiveness of congestion window growth compared to the original H-TCP. This results in slower utilization of available bandwidth.
 - **Stability:** While less aggressive, this modification may lead to improved stability in networks with limited bandwidth or higher congestion, as it avoids sudden bursts in window size

2. UpdateAlpha Function

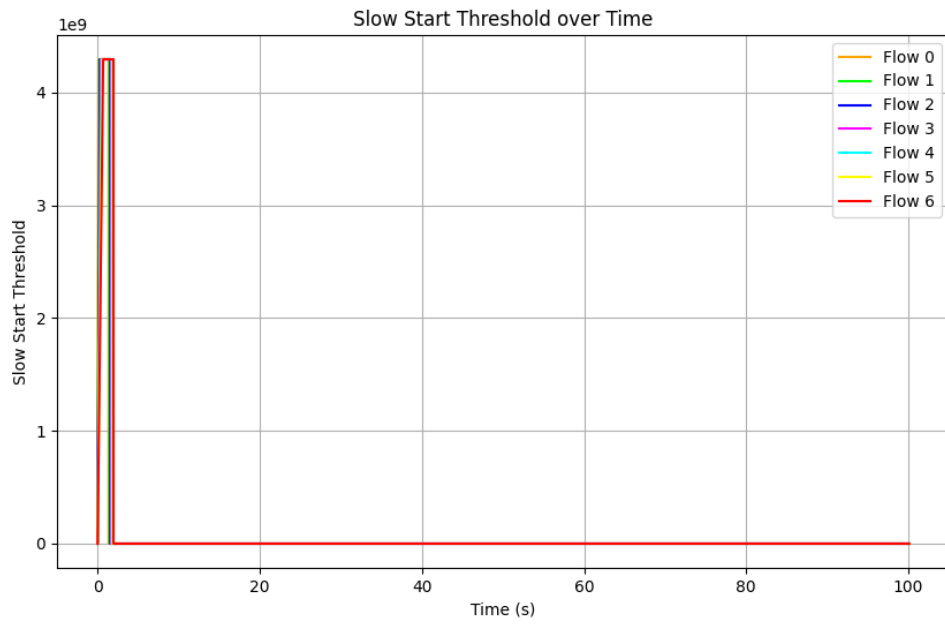
- **Change:**
 - In the original H-TCP, m_alpha grows **quadratically** with respect to the time difference $diffSec$. This makes m_alpha scale very aggressively, especially for larger $diffSec$, allowing faster congestion window growth
 - m_alpha now grows with a **power of 0.8**, which is sub-linear (less than quadratic).
 - The scaling factor **0.6** also reduces the magnitude of growth
- **Potential Impact:**
 - **Less Aggressive Scaling:** The slower growth of m_alpha reduces the rate at which the congestion window can expand during congestion avoidance.
 - **Better Congestion Control:** This modification makes H-TCP more conservative, which may reduce packet loss and congestion in networks with fluctuating RTT or limited resources.
 - **Reduced Throughput in High-Bandwidth Scenarios:** On high-capacity links, the slower scaling of m_alpha may result in underutilization of available bandwidth compared to the original H-TCP.

Congestion Window



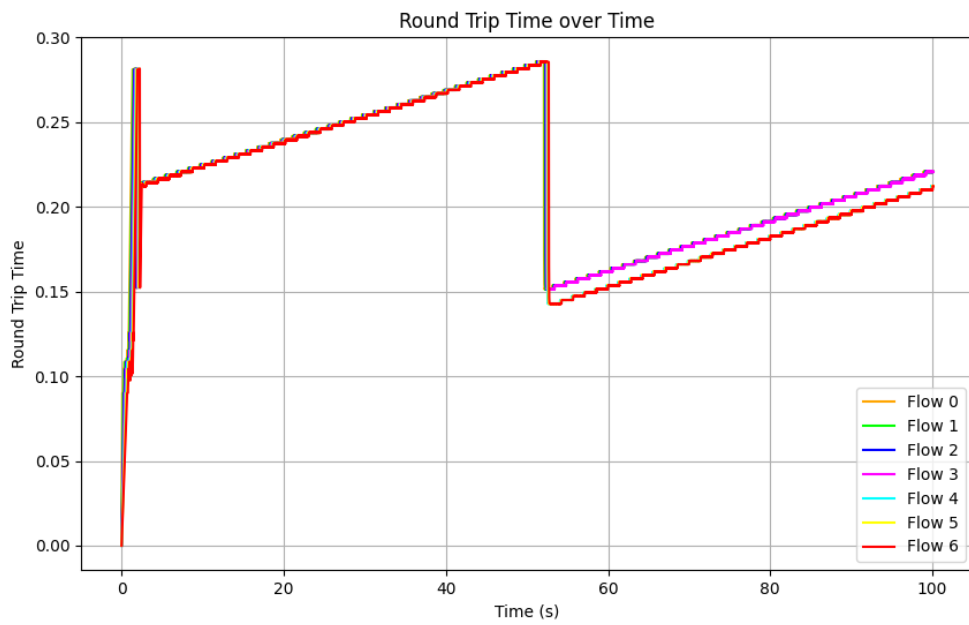
- **The linear scaling reduces the aggressiveness of congestion window growth compared to the original H-TCP**
- **The slower growth of m_{α} reduces the rate at which the congestion window can expand during congestion avoidance**

Slow Start Threshold



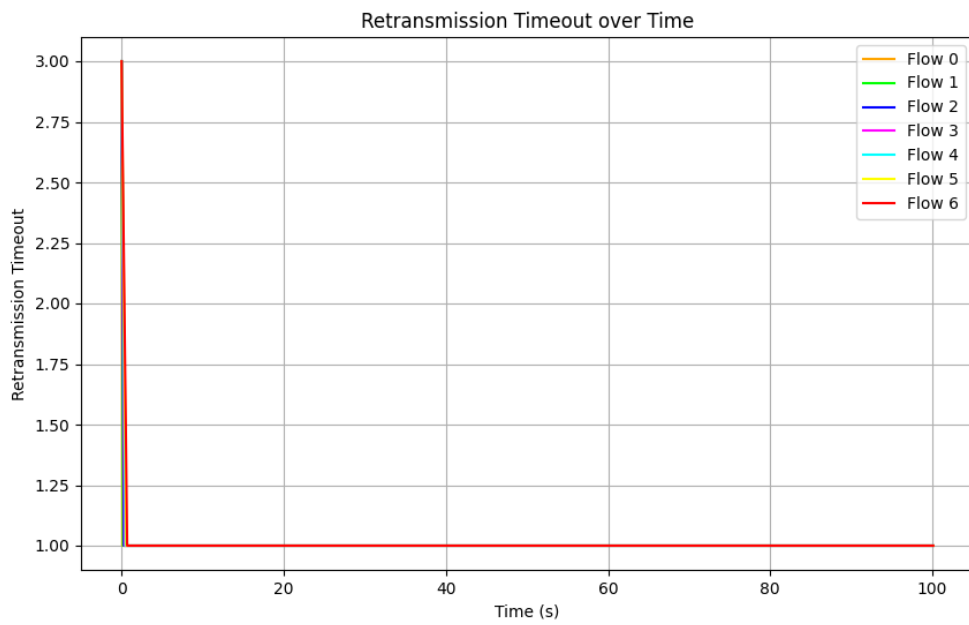
- Exactly same as the original Htcp variant.

Round Trip Time



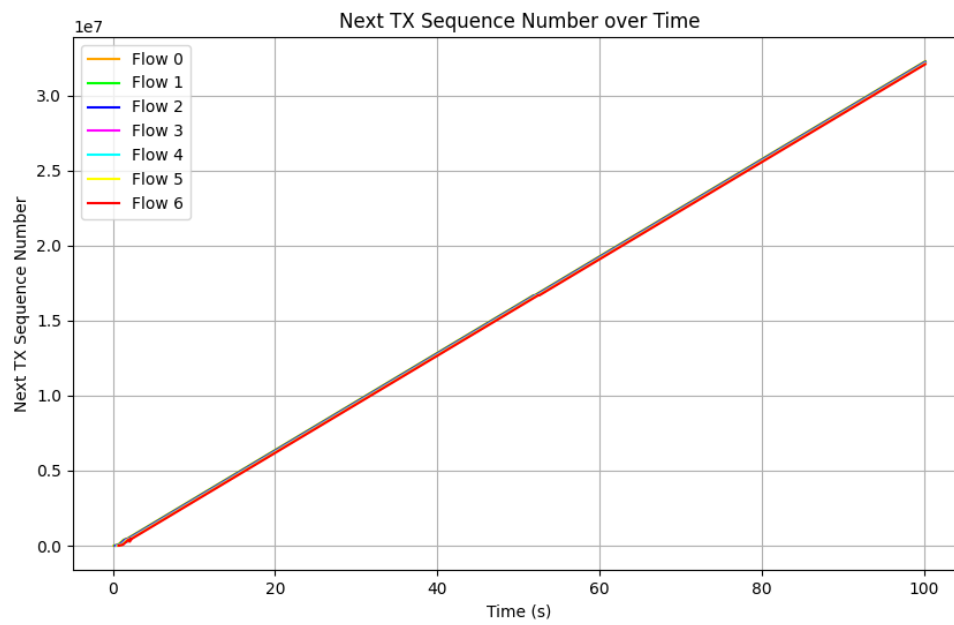
- **The linear scaling reduces the aggressiveness of congestion window growth compared to the original H-TCP**

Retransmission Timeout



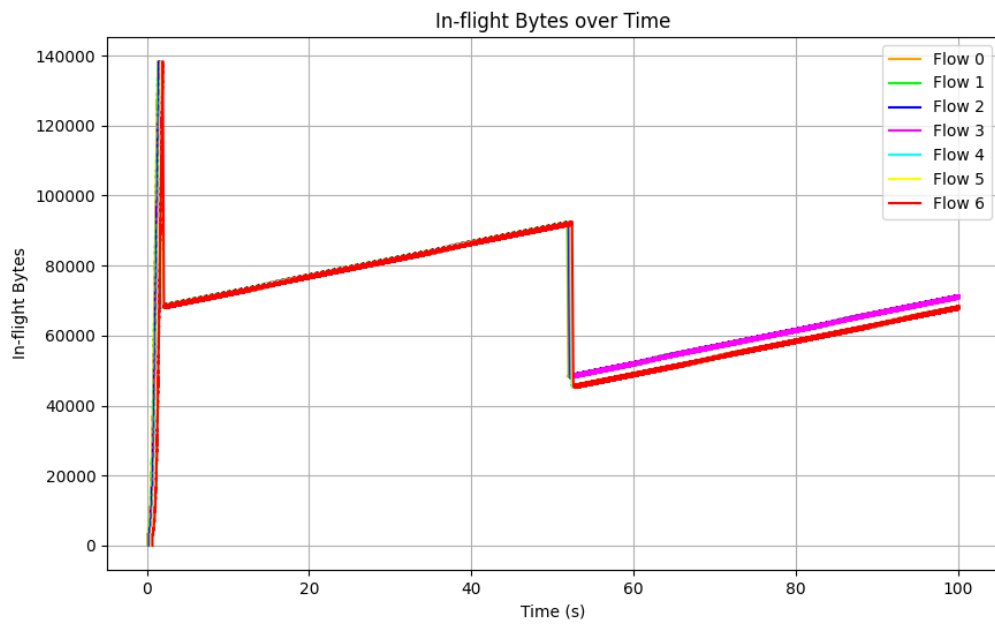
- Exactly same as the original Htcp variant.

Next TX Sequence Number



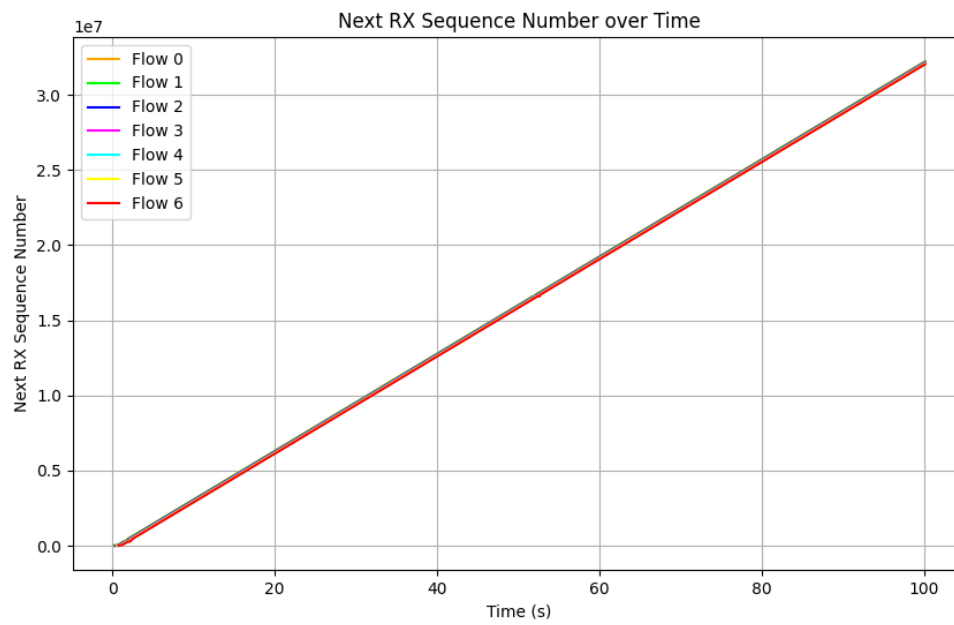
- Exactly same as the original Htcp variant.

In-flight Bytes



- The linear scaling reduces the aggressiveness of congestion window growth compared to the original H-TCP

Next RX Sequence Number



- Exactly same as the original Htcp variant.